WMCS045-05 2026
Lab Assignment #2
Deadline: 2026-09-16 **Group number: _____**

1. Compiler Optimisations

Assume the following code to be stored in a file `main.c`:

```c
#include <stdio.h>

extern long x;

void func();

int main() {
    x = 1;
    func();
    x += 3;

    printf("%ld\n", x);
}
```

Furthermore, assume the following code to be stored in a file `func.c`:

```c
long x;

void func() {
    x += 1;
}
```

Address the following questions and tasks:

a) Which compiler optimisations cannot be performed unless the code is merged into a single file?

b) Check the compiler reports of either `gcc` or `clang`.

c) Merge the code, compare the assembler generated by the compiler when using the optimisation level `-O0` and `-O3`, and check whether the expected optimisations have been performed.

2. Stream Benchmark Suite

Consider the STREAM benchmark suite: `https://github.com/jeffhammond/STREAM`. Use it to explore the memory bandwidth on a system of your choice. Provide the following documentation:

a) Central values and variability of the results as a function of the array length

b) Justification of the parameter choice

c) Benchmarked system

d) Analysis of the results and comparison with the attainable performance

e) Any code modifications

Some hints in this context:

1. Typically, information about the installed memory is available in the BIOS. On a Linux system, the command `dmidecode` can be used to read the BIOS. See, "dmidecode command in Linux with Examples" (GeeksforGeeks, 2024) for more information. Note that `dmidecoce` requires root privileges.

2. Each Kunpeng processor on the HAICGU cluster features 8 memory channels populated with DDR4-2666 memory resulting in a theoretical peak bandwidth of 170.6 GB/s per socket.